Multivariable Calculus

One equation from Chapter 0.1, with a vector in it. Everything else is consequences.

WHERE WE ARE

This entire chapter is one substitution. Take the equation that Chapter 0.1 built the book on,

$$f(a+h) \;=\; f(a) + f'(a)\,h + o(h),$$

and let h be a *vector* and $f'(a)$ a *linear map* (Chapter 0.4). That is it. That is the whole idea. Partial derivatives, the gradient, the chain rule as a product of matrices, the Hessian, Lagrange multipliers and the Jacobian determinant are all consequences of that one move, and this chapter derives them in that order.

This is the promised payoff for a decision made on the first page. Chapter 0.1 insisted, at some cost in apparent pedantry, that the derivative is **the coefficient of the best linear approximation** and not a slope. Had we defined it as a slope, we would now be stuck: there is no such thing as "the slope" of a function of three variables — there are infinitely many, one per direction, and no way to choose. There *is* exactly one best linear approximation. The general definition costs nothing extra here; the special one would have cost us the chapter.

One section — §4 — is more important than it looks. It observes that the gradient is not really a vector, that turning it into one requires an inner product you did not notice yourself using, and that this stops being invisible the moment you leave Cartesian coordinates. You will meet your first covector there. Chapters 2.4, 3.2 and 3.3 are built on it.

TOOLS YOU'LL NEED — Chapter 0.1: the derivative as a linear approximation, the $o(h)$ bookkeeping, the chain rule as composition. Chapter 0.3: Taylor's theorem with remainder, used once in §6. Chapter 0.4: linear maps, their matrices in a basis, composition as matrix multiplication, and the determinant as a signed volume scaling factor. Chapter 0.5: inner products, Cauchy–Schwarz with its equality condition, orthogonal projection, and the spectral theorem for symmetric matrices. Chapters 0.4 and 0.5 are not decoration here — §3, §6 and §7 each cash a specific theorem from them.

1 · Partial derivatives, and the trap

The obvious first move, when a function has several inputs, is to freeze all but one and differentiate in the survivor. Let $f : \mathbb{R}^n \to \mathbb{R}$ and let $\mathbf{e}_1, \ldots, \mathbf{e}_n$ be the standard basis. The **partial derivative** of f with respect to x^i at $\mathbf{a}$ is

$$\frac{\partial f}{\partial x^i}(\mathbf{a}) \;\equiv\; \partial_i f(\mathbf{a}) \;=\; \lim_{t\to 0} \frac{f(\mathbf{a}+t\,\mathbf{e}_i) - f(\mathbf{a})}{t}. \tag{0.6.1}$$

Nothing new is happening. Restricted to the line $\mathbf{a} + t\,\mathbf{e}_i$, the function f is a function of the single variable t, and (0.6.1) is Chapter 0.1's definition applied to it. Every rule you already have — product, quotient, chain — applies verbatim, because during the computation the other variables are literally constants. So $\partial_x(x^2y^3) = 2xy^3$ and $\partial_y(x^2y^3) = 3x^2y^2$, and there is no technique to learn.

The trap is that this feels like it ought to be the whole story, and it is not. Partial derivatives are a genuinely *weaker* notion than differentiability, and the gap is not subtle.

THE COUNTEREXAMPLE TO HAVE PERMANENTLY

Define

$$f(x,y) = \frac{xy}{x^2+y^2} \quad \text{for } (x,y) \neq (0,0), \qquad f(0,0) = 0.$$

Both partial derivatives exist at the origin, and both are zero. But f is *not continuous* at the origin — it does not even have a limit there.

Check the partials first. Along the x-axis, $f(x,0) = 0/(x^2) = 0$ for every x, so the function $t \mapsto f(t,0)$ is identically zero and its derivative at $t = 0$ is 0. Hence $\partial_x f(0,0) = 0$, and by the symmetry $f(x,y) = f(y,x)$ also $\partial_y f(0,0) = 0$. Both partials exist, are finite, and are perfectly well behaved.

Now approach the origin along the diagonal $y = x$ instead. For $x \neq 0$,

$$f(x,x) \;=\; \frac{x \cdot x}{x^2 + x^2} \;=\; \frac{x^2}{2x^2} \;=\; \tfrac{1}{2}, \tag{0.6.2}$$

a constant. So $f \to \frac{1}{2}$ along the diagonal and $f \to 0$ along the axes. Two different limits, so there is no limit, so f is not continuous at the origin — and a function that is not continuous at a point certainly does not have a good linear approximation there.

It is worth seeing *why* the function is so badly behaved, because the reason is geometrical. Write $x = r\cos\theta, y = r\sin\theta$. Then

$$f \;=\; \frac{r\cos\theta \cdot r\sin\theta}{r^2} \;=\; \cos\theta\sin\theta \;=\; \tfrac{1}{2}\sin 2\theta. \tag{0.6.3}$$

The radius has cancelled completely: f depends only on the *direction* of approach, never on the distance. Every value in $[-\frac{1}{2}, \frac{1}{2}]$ is attained arbitrarily close to the origin, on some ray. The two coordinate axes happen to be rays on which f vanishes, and the partial derivatives look along exactly those two rays and see nothing wrong. They are two samples from a circle's worth of behaviour.

That is the honest indictment. **Partial derivatives probe n lines through a point; a function has infinitely many directions and can misbehave in all the ones you didn't check.** So we need a definition that constrains the function's behaviour in *every* direction at once, and constrains it uniformly. Chapter 0.1 already wrote it down; we only have to let the displacement be a vector.

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2 · The total derivative

Here is the definition the rest of the chapter runs on. Let $f : \mathbb{R}^n \to \mathbb{R}^m$.

> **DEFINITION — THE TOTAL DERIVATIVE**
>
> f is **differentiable at a** if there exists a *linear map* $\mathrm{D}f(\mathbf{a}) : \mathbb{R}^n \to \mathbb{R}^m$ such that
>
> $$f(\mathbf{a}+\mathbf{h}) \;=\; f(\mathbf{a}) \;+\; \mathrm{D}f(\mathbf{a})\,\mathbf{h} \;+\; o(|\mathbf{h}|), \tag{0.6.4}$$
>
> where, exactly as in Chapter 0.1, "$o(|\mathbf{h}|)$" means a remainder $R(\mathbf{h})$ with $|R(\mathbf{h})| \,/\, |\mathbf{h}| \to 0$ as $\mathbf{h} \to \mathbf{0}$. The map $\mathrm{D}f(\mathbf{a})$ is the **total derivative**, or the **differential**, of f at $\mathbf{a}$.

Compare it with the one-variable equation letter by letter. a became $\mathbf{a}$; h became $\mathbf{h}$; the *number* $f'(a)$ became the *linear map* $\mathrm{D}f(\mathbf{a})$; and "multiply by $f'(a)$" became "apply the linear map". Multiplication by a number is what a linear map $\mathbb{R} \to \mathbb{R}$ does, so the old definition is the case $n = m = 1$ of the new one, not an analogy to it.

Notice how much stronger this is than §1. The limit is over $\mathbf{h} \to \mathbf{0}$ with no restriction on direction, so the approximation must hold uniformly as $\mathbf{h}$ approaches zero *any way at all* — including along spirals and along the diagonal that broke the counterexample. Differentiability at $\mathbf{a}$ immediately forces continuity at $\mathbf{a}$: as $\mathbf{h} \to \mathbf{0}$ the right-hand side of (0.6.4) tends to $f(\mathbf{a})$, since a linear map sends $\mathbf{0}$ to $\mathbf{0}$ and is continuous.

2.1 · There is only one such map

The definition says "there exists". It is worth two lines to check that it cannot exist twice, because otherwise "*the* derivative" would be an abuse of language.

Suppose A and B both satisfy (0.6.4). Subtracting the two statements, the function values cancel and we are left with $(A-B)\mathbf{h} = o(|\mathbf{h}|)$. Fix any direction $\mathbf{u}$ with $|\mathbf{u}| = 1$ and put $\mathbf{h} = t\mathbf{u}$ for small $t > 0$. By linearity $(A - B)(t\mathbf{u}) = t\,(A-B)\mathbf{u}$, so

$$\frac{|(A-B)(t\mathbf{u})|}{|t\mathbf{u}|} \;=\; \frac{t\,|(A-B)\mathbf{u}|}{t} \;=\; |(A-B)\mathbf{u}|\,. \tag{0.6.5}$$

The left side must tend to 0 as $t \to 0$; the right side does not depend on t at all. A constant that tends to zero is zero, so $(A-B)\mathbf{u} = \mathbf{0}$ for every $\mathbf{u}$, so $A = B$. The derivative, when it exists, is unique — and note the mechanism: *the linearity did all the work*, by letting us pull the small parameter out and cancel it.

2.2 · Its matrix is the Jacobian

$\mathrm{D}f(\mathbf{a})$ is a linear map, so by Chapter 0.4 it has a matrix once we fix bases, and the recipe for that matrix is: *the j-th column is the image of the j-th basis vector*. So feed basis vectors into the definition. Put $\mathbf{h} = t\,\mathbf{e}_j$ in (0.6.4):

$$f(\mathbf{a} + t\,\mathbf{e}_j) \;=\; f(\mathbf{a}) + t\,\mathrm{D}f(\mathbf{a})\mathbf{e}_j + o(t), \tag{0.6.6}$$

using $\mathrm{D}f(\mathbf{a})(t\mathbf{e}_j) = t\,\mathrm{D}f(\mathbf{a})\mathbf{e}_j$ by linearity. Rearrange and divide by t:

$$\frac{f(\mathbf{a} + t\,\mathbf{e}_j) - f(\mathbf{a})}{t} \;=\; \mathrm{D}f(\mathbf{a})\mathbf{e}_j + \frac{o(t)}{t} \;\xrightarrow[t\to 0]{}\; \mathrm{D}f(\mathbf{a})\mathbf{e}_j. \tag{0.6.7}$$

But the left-hand side is, componentwise, exactly the definition (0.6.1) of the partial derivative with respect to x^j. Therefore the j-th column of the matrix of $\mathrm{D}f(\mathbf{a})$ is the column of partials $\partial_j f^1, \ldots, \partial_j f^m$, and the matrix is the **Jacobian**:

$$\mathrm{D}f(\mathbf{a}) \;\longleftrightarrow\; \begin{pmatrix} \partial_1 f^1 & \cdots & \partial_n f^1 \\ \vdots & \ddots & \vdots \\ \partial_1 f^m & \cdots & \partial_n f^m \end{pmatrix}, \qquad [\mathrm{D}f]^i{}_j = \frac{\partial f^i}{\partial x^j}. \tag{0.6.8}$$

Read the shape: m rows, one per output component; n columns, one per input direction. A map $\mathbb{R}^3 \to \mathbb{R}^2$ has a 2×3 Jacobian, and there is never any doubt about which way round it goes, because a matrix acting on a column vector in $\mathbb{R}^n$ must have n columns.

So the partials do reconstruct the derivative — *once you know the derivative exists*. That qualifier is the entire content of §1. The logic runs one way only:

$$\text{differentiable at } \mathbf{a} \;\Longrightarrow\; \text{all partials exist at } \mathbf{a}, \tag{0.6.9}$$

and the converse is false, by (0.6.3). What rescues the converse is one extra hypothesis, and this is the one theorem in the chapter we will quote rather than prove in the main text:

> **THEOREM (QUOTED — PROVED IN THE GRIND BOX BELOW)**
>
> If all the partial derivatives $\partial_j f^i$ exist in a neighbourhood of $\mathbf{a}$ and are **continuous at $\mathbf{a}$**, then f is differentiable at $\mathbf{a}$, and $\mathrm{D}f(\mathbf{a})$ is the Jacobian (0.6.8).

Why it is safe to lean on this: every function you will differentiate in this book is built from polynomials, exponentials, logarithms and trigonometric functions by arithmetic and composition, and the partials of such a thing are again of the same type and continuous wherever they are defined. So in practice you compute the partials, observe that they are continuous, and conclude that the Jacobian really is the derivative. The counterexample of §1 is not a warning about your daily calculations; it is a warning about what the *definition* has to be if the theorems are going to be true. We are quoting the theorem rather than deriving it in the flow of the argument only because its proof needs the mean value theorem from Chapter 0.2 and adds nothing to the picture — but it is written out below, because this book does not say "it can be shown".

▼ **Grind box — the boundedness lemma, and continuous partials ⇒ differentiable**

Lemma (linear maps are bounded). For any linear $A : \mathbb{R}^n \to \mathbb{R}^m$ there is a constant C with $|A\mathbf{v}| \le C\,|\mathbf{v}|$ for all $\mathbf{v}$.

Proof. Write $\mathbf{v} = \sum_i v^i \mathbf{e}_i$, so $A\mathbf{v} = \sum_i v^i A\mathbf{e}_i$. By the triangle inequality and then Cauchy–Schwarz (Chapter 0.5) applied to the two lists of numbers $\left|v^i\right|$ and $|A\mathbf{e}_i|$,

$$|A\mathbf{v}| \;\le\; \sum_i \left|v^i\right|\,|A\mathbf{e}_i| \;\le\; \left(\sum_i (v^i)^2\right)^{1/2} \left(\sum_i |A\mathbf{e}_i|^2\right)^{1/2} \;=\; C\,|\mathbf{v}|$$

with $C = \left(\sum_i |A\mathbf{e}_i|^2\right)^{1/2}$, a number depending only on A. ∎

This innocuous fact is used constantly and silently: it is what lets us say $\mathrm{D}f\left(o(|\mathbf{h}|)\right) = o(|\mathbf{h}|)$, i.e. that a linear map does not turn a negligible displacement into a non-negligible one. It is also where finite-dimensionality enters. In infinite dimensions (Chapter 4.3) linear maps can be unbounded, $\frac{\mathrm{d}}{\mathrm{d}x}$ being the standard offender, and a great deal of quantum mechanics' technical difficulty descends from exactly that.

Theorem. If the partials of f exist near $\mathbf{a}$ and are continuous at $\mathbf{a}$, then f is differentiable at $\mathbf{a}$.

Proof for $n = 2$, $m = 1$; the general case is the same argument with more terms. Write $\mathbf{a} = (a, b)$ and $\mathbf{h} = (h, k)$. Split the total change into two moves, one along each axis, by adding and subtracting the corner value:

$$f(a+h, b+k) - f(a,b) = \underbrace{\big[f(a+h, b+k) - f(a, b+k)\big]}_{\text{move in } x} + \underbrace{\big[f(a, b+k) - f(a,b)\big]}_{\text{move in } y}.$$

Each bracket changes only one variable, so each is a one-variable increment and the mean value theorem of Chapter 0.2 applies to it. There exist $\theta_1, \theta_2 \in (0,1)$ with

$$f(a+h, b+k) - f(a,b) = h\,\partial_x f(a+\theta_1 h,\; b+k) \;+\; k\,\partial_y f(a,\; b+\theta_2 k).$$

Now use continuity of the partials *at* (a,b): both evaluation points converge to (a,b) as $(h,k) \to (0,0)$, so we may write $\partial_x f(a+\theta_1 h, b+k) = \partial_x f(a,b) + \varepsilon_1$ and $\partial_y f(a, b+\theta_2 k) = \partial_y f(a,b) + \varepsilon_2$ with $\varepsilon_1, \varepsilon_2 \to 0$. Then

$$f(\mathbf{a}+\mathbf{h}) - f(\mathbf{a}) = \underbrace{h\,\partial_x f(\mathbf{a}) + k\,\partial_y f(\mathbf{a})}_{\text{linear in } \mathbf{h}} \;+\; \underbrace{h\varepsilon_1 + k\varepsilon_2}_{\text{remainder}},$$

and since $|h| \le |\mathbf{h}|$ and $|k| \le |\mathbf{h}|$,

$$\frac{|h\varepsilon_1 + k\varepsilon_2|}{|\mathbf{h}|} \le |\varepsilon_1| + |\varepsilon_2| \longrightarrow 0.$$

That is exactly (0.6.4) with the Jacobian as the linear map. ∎

Where the counterexample of §1 fails this test: its partials do exist away from the origin, but a short computation from (0.6.3) shows they blow up like $1/r$ as you approach it, so they are nowhere near continuous at the origin. The hypothesis is not decorative.

⚠ **WHY THIS ISN'T OBVIOUS**

The symbol $\frac{\partial f}{\partial x}$ hides a choice, and the notation gives you no warning. In one variable there is nothing to hold fixed, so $\frac{\mathrm{d}f}{\mathrm{d}x}$ is unambiguous. In several variables the phrase "differentiate with respect to x" is incomplete until you say *along which direction*, which is the same as saying *what is being held constant* — and the symbol does not say.

Here is the cleanest possible demonstration, with no physics in it. Take the function $f = y$ on the plane, and describe the plane in two ways: by coordinates (x, y), and by coordinates (x, z) where $z = x + y$. Both are legitimate coordinate systems; both contain a coordinate called x. Then

$$\left(\frac{\partial f}{\partial x}\right)_y = \frac{\partial}{\partial x} y = 0, \qquad \text{while} \qquad \left(\frac{\partial f}{\partial x}\right)_z = \frac{\partial}{\partial x}(z - x) = -1.$$

Same function, same symbol $\partial/\partial x$, answers 0 and -1. Nothing has gone wrong: the two derivatives are taken along different directions — one along the line $y =$ const, the other along the line $z =$ const — and there is no reason for them to agree. The subscript is not fussiness; without it the expression is not defined.

This is why thermodynamics is written the way it is. From the first law $\mathrm{d}U = T\,\mathrm{d}S - p\,\mathrm{d}V$, holding S fixed gives $\left(\frac{\partial U}{\partial V}\right)_S = -p$. But for an ideal gas the internal energy depends only on temperature — Joule's free-expansion experiment — so $\left(\frac{\partial U}{\partial V}\right)_T = 0$. For air at atmospheric pressure the first is about $-10^5\ \mathrm{J\,m^{-3}}$ and the second is exactly zero. Two genuinely different numbers wearing the same-looking symbol, and generations of students have lost marks to the difference.

Now notice what the total derivative does to this problem: **it dissolves it.** $\mathrm{D}f(\mathbf{a})$ is a single object that eats *all* displacement vectors at once and returns the corresponding first-order change. It does not privilege any direction, so there is nothing to hold fixed and nothing to suppress. The ambiguity was never in the mathematics; it was in the decision to decompose a directionless object along a basis and then forget which basis. That is a second, independent argument — the first was §1 — for treating $\mathrm{D}f$ rather than the partials as the fundamental object. The partials are its components, and components always presuppose a basis.

3 · The gradient

Specialise to a scalar-valued function, $m = 1$, which is the case physics uses most: a temperature field, a potential, an action, a log-likelihood. Now $\mathrm{D}f(\mathbf{a}) : \mathbb{R}^n \to \mathbb{R}$ is a linear map into the reals — a *linear functional*, in Chapter 0.4's terms a linear map whose target happens to be one-dimensional. Its matrix is therefore a single row, $\begin{pmatrix}\partial_1 f & \cdots & \partial_n f\end{pmatrix}$.

A row of numbers looks like a vector lying on its side, and the temptation to stand it up is irresistible. Give in to it — but notice the step, because §4 is about what it costs. Using the standard inner product of Chapter 0.5, define the **gradient** $\nabla f(\mathbf{a})$ as the unique vector satisfying

$$\mathrm{D}f(\mathbf{a})\,\mathbf{h} \;=\; \nabla f(\mathbf{a}) \cdot \mathbf{h} \qquad \text{for every } \mathbf{h}. \tag{0.6.10}$$

Its components follow by feeding in basis vectors, exactly as in §2.2: taking $\mathbf{h} = \mathbf{e}_i$ gives $\nabla f \cdot \mathbf{e}_i = \partial_i f$, and the left side is the i-th component of ∇f. So

$$\nabla f \;=\; \big(\partial_1 f,\ \partial_2 f,\ \ldots,\ \partial_n f\big). \tag{0.6.11}$$

3.1 · Steepest ascent, derived from Cauchy–Schwarz

Ask the natural question: in which direction does f increase fastest? Make it precise. For a unit vector $\hat{u}$, the **directional derivative** is the rate of change of f along $\hat{u}$,

$$D_{\hat{u}} f(\mathbf{a}) \;=\; \lim_{t \to 0} \frac{f(\mathbf{a} + t\hat{u}) - f(\mathbf{a})}{t} \;=\; \mathrm{D}f(\mathbf{a})\hat{u} \;=\; \nabla f \cdot \hat{u}, \tag{0.6.12}$$

where the middle equality is (0.6.4) with $\mathbf{h} = t\hat{u}$, divided by t, in the limit — the same computation as (0.6.7) but with an arbitrary direction instead of a basis vector. So the question is: over all unit $\hat{u}$, which maximises $\nabla f \cdot \hat{u}$?

That is precisely the question Cauchy–Schwarz answers. Chapter 0.5 proved that for any two vectors,

$$|\mathbf{p} \cdot \mathbf{q}| \;\le\; |\mathbf{p}|\;|\mathbf{q}|\,, \qquad \text{with equality iff } \mathbf{p} \parallel \mathbf{q}. \tag{0.6.13}$$

Apply it with $\mathbf{p} = \nabla f$ and $\mathbf{q} = \hat{u}$, remembering $|\hat{u}| = 1$:

$$-\,|\nabla f| \;\le\; \nabla f \cdot \hat{u} \;\le\; +\,|\nabla f|\,. \tag{0.6.14}$$

Both bounds are attained, by the equality clause: the upper one at $\hat{u} = \nabla f / |\nabla f|$, where $\nabla f \cdot \hat{u} = |\nabla f|^2 / |\nabla f| = |\nabla f|$; the lower one at $\hat{u} = -\nabla f / |\nabla f|$. Hence, with nothing assumed and nothing waved at:

STEEPEST ASCENT

The direction of fastest increase of f at $\mathbf{a}$ is $\nabla f / |\nabla f|$; the rate of increase in that direction is $|\nabla f|$; the direction of fastest decrease is the opposite one, with rate $-|\nabla f|$. The gradient does not merely *point* uphill — its length *is* the steepness.

This is the cleanest single payoff for having proved Cauchy–Schwarz. Note also what happens at the other extreme: if $\nabla f \cdot \hat{u} = 0$ then f does not change at all to first order along $\hat{u}$. Those directions are worth a name, and they have a geometric meaning.

3.2 · The gradient is perpendicular to level sets

A **level set** of f is a set $\{\mathbf{x} : f(\mathbf{x}) = c\}$ — an isotherm, an equipotential, a contour line on a map. Let $\mathbf{x}(t)$ be any differentiable curve that lies inside one of them, so that $f(\mathbf{x}(t)) = c$ for all t. Differentiate that identity.

We can do this from the definition, which also quietly proves the special case of the chain rule we will need repeatedly. Expand $\mathbf{x}$ to first order, $\mathbf{x}(t+s) = \mathbf{x}(t) + s\,\dot{\mathbf{x}}(t) + o(s)$, and feed the result into (0.6.4):

$$\begin{aligned} f\big(\mathbf{x}(t+s)\big) &= f\big(\mathbf{x}(t)\big) + \mathrm{D}f\big[s\,\dot{\mathbf{x}}(t) + o(s)\big] + o(s) \\ &= f\big(\mathbf{x}(t)\big) + s\,\nabla f \cdot \dot{\mathbf{x}}(t) + o(s), \end{aligned} \tag{0.6.15}$$

where the boundedness lemma of §2 was used to absorb $\mathrm{D}f\big[o(s)\big]$ into $o(s)$. Comparing with the one-variable definition of a derivative in s, we read off

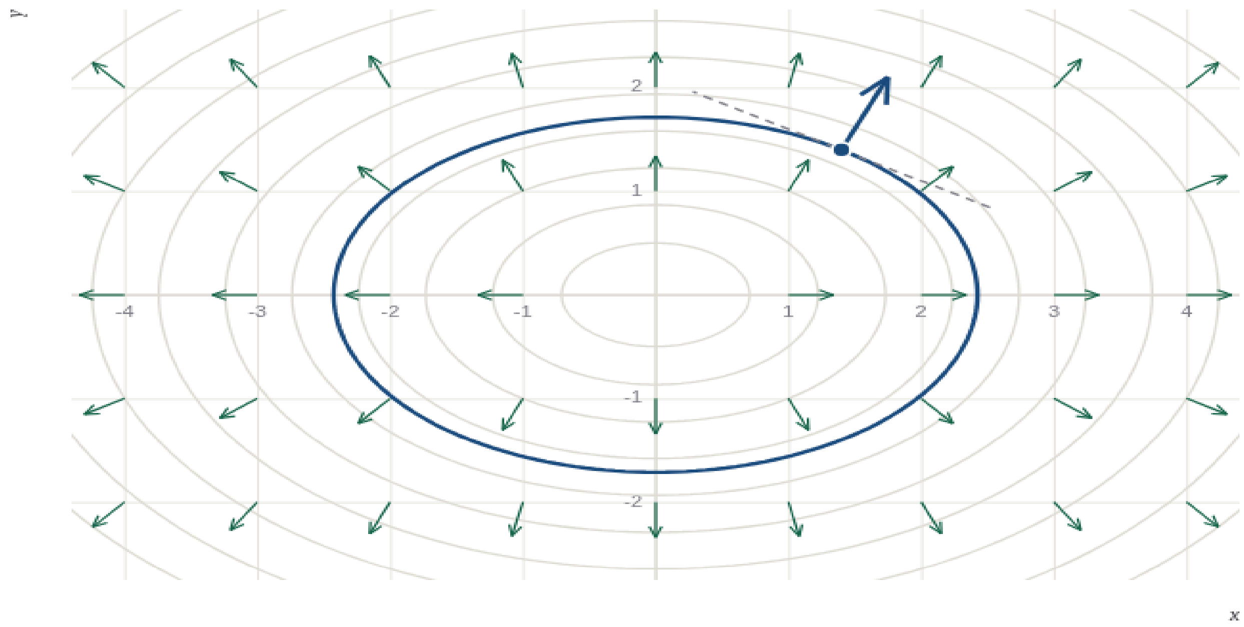
$$\frac{\mathrm{d}}{\mathrm{d}t} f\big(\mathbf{x}(t)\big) \;=\; \nabla f \cdot \dot{\mathbf{x}}(t). \tag{0.6.16}$$

Now impose that the curve lies in a level set: the left side is $\frac{\mathrm{d}c}{\mathrm{d}t} = 0$, so

$$\nabla f \cdot \dot{\mathbf{x}} \;=\; 0 \qquad \text{for every curve inside a level set.} \tag{0.6.17}$$

The velocity $\dot{\mathbf{x}}$ of such a curve is exactly what we mean by a vector *tangent* to the level set. So ∇f is orthogonal to every tangent direction of the level set through $\mathbf{a}$: **the gradient is perpendicular to the level surface, and points across it in the direction of increase.**

The two facts fit together into one picture. Contours are lines of no change; the gradient is perpendicular to them; and it is longest where the contours are closest together, because that is where a small step changes f the most. The figure below is that picture, with the arithmetic exposed so you can check it rather than believe it.



point *x* point *y* **show constraint (§7)**

`∇f = (2.80, 5.60), |∇f| = 6.261    ∇f · t̂ = -2.8e-12 (zero)`

One figure, three sections. Grey ellipses are level sets of $f(x, y) = x^2 + 2y^2$; green arrows are the gradient field, all drawn the same length so that only their *direction* is being claimed. Drag the point (or use the sliders). The blue curve is the level set through it, the dashed grey line is that curve's tangent, and the thick blue arrow is ∇f there, drawn to scale. The readout reports $\nabla f \cdot \hat{t}$, computed by taking a numerical tangent from the contour and dotting it with the analytic gradient: it stays around 10^{-12}, which is machine roundoff and nothing else — (0.6.17) measured rather than asserted. Watch the blue arrow lengthen as you move outward, where the contours crowd — that is §3.1's claim that $|\nabla f|$ *is* the steepness — and shrink to nothing at the origin, where the level sets degenerate to a point and the gradient vanishes. **Now press the button.** The orange line is the constraint $g(x, y) = x + y = 3$. Its lowest-f point is marked, and there the purple level curve of f is *tangent* to the constraint line, so the two gradients are parallel: $\nabla f = (4, 4) = 4\,\nabla g$. That number 4 is the Lagrange multiplier of §7, and it is also $\frac{\mathrm{d}f^*}{\mathrm{d}c}$. Slide the point along the orange line and watch the blue contour cut across it everywhere except at the marked point — which is the whole proof of §7 in one image.

4 · The gradient is really a one-form — and why you must care

Go back to (0.6.10) and look at what was actually done there. We had a perfectly good object, the linear map $\mathrm{D}f(\mathbf{a})$, which eats a displacement vector and returns a number. We converted it into a *vector*, ∇f, by demanding that dotting with that vector reproduce the map. The conversion required a dot product. It could not have been done without one.

Give the natural object its usual name. Write

$$\mathrm{d}f(\mathbf{h}) \;\equiv\; \mathrm{D}f(\mathbf{a})\,\mathbf{h} \;=\; h^i\,\partial_i f, \qquad (0.6.18)$$

a sum over i. The object $\mathrm{d}f$ is a **one-form**, or **covector**: a linear machine that eats a vector and returns a number. In Chapter 0.4's language it is nothing but a linear map whose target happens to be one-dimensional; and since such maps can be added and scaled and the result is another one, they form a vector space in their own right — the **dual space** V^*, which Chapter 2.4 will treat properly. What matters here is that $\mathrm{d}f$ is defined by f alone. No inner product, no notion of length, no notion of angle went into (0.6.18) — only the pairing of a displacement with a rate of change, which is exactly what a physical derivative is. (This is also the honest definition of the $\mathrm{d}x$ you have been cancelling since school; Chapter 0.1's warning callout promised you it would get one here.)

The gradient, by contrast, is defined by *two* things: f and an inner product. Look at the index positions, which are about to earn their keep. In (0.6.18) the components h^i of a vector carry an **upper** index; the components $\partial_i f$ of the one-form carry a **lower** one; and they are summed against each other to make a number.

Now bring in a general inner product. Given a basis, bilinearity determines it completely by what it does to basis vectors: writing $\mathbf{p} = p^i\mathbf{e}_i$ and $\mathbf{q} = q^j\mathbf{e}_j$ and expanding,

$$\mathbf{p}\cdot\mathbf{q} \;=\; p^i q^j \left(\mathbf{e}_i\cdot\mathbf{e}_j\right) \;\equiv\; g_{ij}\,p^i q^j, \qquad g_{ij} \equiv \mathbf{e}_i\cdot\mathbf{e}_j. \tag{0.6.19}$$

The array g_{ij} is called the **metric** in that basis. It is symmetric because the inner product is, and it is invertible because the inner product is non-degenerate — if some $\mathbf{v}$ had $g_{ij}v^i = 0$ for all j then $\mathbf{v}$ would be orthogonal to everything, including itself, so $\mathbf{v} = \mathbf{0}$. Both index positions are down, matching the fact that it eats two vectors. Write the defining relation (0.6.10) in these components:

$$g_{ij}\,(\nabla f)^i\, h^j \;=\; \partial_j f\; h^j \qquad \text{for every } \mathbf{h}, \tag{0.6.20}$$

so the coefficients of each h^j must agree: $g_{ij}(\nabla f)^i = \partial_j f$. Multiplying by the inverse matrix g^{ij} (which exists because the inner product is non-degenerate),

$$\boxed{(\nabla f)^i \;=\; g^{ij}\,\partial_j f.} \tag{0.6.21}$$

There it is, in the open. **Converting the one-form $\partial_j f$ into the vector $(\nabla f)^i$ is exactly the act of contracting with the inverse metric.** It is called *raising an index*, and it is not free.

In Cartesian coordinates on Euclidean space $g_{ij} = \delta_{ij}$, so $g^{ij} = \delta^{ij}$ and (0.6.21) reads $(\nabla f)^i = \partial_i f$: the components are numerically identical and the distinction is invisible. That invisibility is why the gradient is usually taught as a vector and why the whole issue can seem like pedantry. It is not pedantry, and here is where it stops being invisible:

- **Non-Cartesian coordinates, immediately.** In plane polar coordinates the metric is $g_{ij} = \mathrm{diag}(1, r^2)$, so $g^{ij} = \mathrm{diag}(1, r^{-2})$ and the gradient's angular component is $r^{-2}\partial_\theta f$, not $\partial_\theta f$. If you have ever wondered where the stray $1/r$ in the polar gradient formula comes from, it comes from here — see the grind box.
- **Relativity, permanently.** The Minkowski metric has signature $(+,-,-,-)$ (Chapter 2.3), so raising a spatial index *flips its sign*: $(\nabla f)^i = -\partial_i f$. There is no coordinate choice that makes this go away, because no coordinate choice makes the metric the identity. From Chapter 2.4 onward, upper and lower indices are different objects and mixing them is an error, not a style preference.

- **Curved space, structurally.** On a manifold (Chapter 3.2) there is no inner product at all until you supply one — and supplying one *is* the gravitational field (Chapter 3.3). So on a manifold $\mathrm{d}f$ exists always and ∇f exists only once gravity has been specified. That is a statement about physics, not notation.

So: you have just met your first covector, and it was hiding in the most familiar object in vector calculus. The rule to carry forward is short. **Differentiation naturally produces lower indices; vectors naturally carry upper ones; the metric is what converts between them, and when the metric is the identity you cannot see the conversion happening.** Chapter 2.4 makes this a formal convention and Chapter 3.2 makes it unavoidable. Nothing more is needed here.

▼ **Grind box — the polar gradient, derived rather than memorised**

In plane polar coordinates (r, θ) the squared length of a small displacement is, by Pythagoras on a radial step $\mathrm{d}r$ and a perpendicular arc $r\,\mathrm{d}\theta$,

$$\mathrm{d}s^2 = \mathrm{d}r^2 + r^2\mathrm{d}\theta^2 \qquad \Longrightarrow \qquad g_{ij} = \begin{pmatrix} 1 & 0 \\ 0 & r^2 \end{pmatrix}.$$

Inverting a diagonal matrix is easy: $g^{ij} = \mathrm{diag}(1,\, r^{-2})$. So by (0.6.21) the gradient's components in the coordinate basis $\{\partial_r, \partial_\theta\}$ are

$$(\nabla f)^r = \partial_r f, \qquad (\nabla f)^\theta = \frac{1}{r^2}\,\partial_\theta f.$$

The coordinate basis vector ∂_θ is not a unit vector — moving one unit of θ moves you a distance r — so its normalised version is $\hat{\boldsymbol{\theta}} = r^{-1}\partial_\theta$, i.e. $\partial_\theta = r\,\hat{\boldsymbol{\theta}}$. Substituting,

$$\nabla f = (\nabla f)^r\,\hat{\mathbf{r}} + (\nabla f)^\theta\, r\,\hat{\boldsymbol{\theta}} = \frac{\partial f}{\partial r}\,\hat{\mathbf{r}} + \frac{1}{r}\frac{\partial f}{\partial \theta}\,\hat{\boldsymbol{\theta}}.$$

That is the formula printed on the inside cover of every vector-calculus text, and you have just derived it. Two morals. First, the notorious $1/r$ is not a quirk of polar coordinates; it is *one* factor of r^{-2} from raising the index and *one* factor of r from normalising the basis vector. Second — and this is the part that matters later — the "h_i scale factors" that vector-calculus books ask you to memorise for cylindrical and spherical coordinates are nothing but the diagonal entries $\sqrt{g_{ii}}$ of the metric. Chapter 3.3 replaces the table with the metric and never mentions scale factors again.

Sanity check: for $f = r$ we get $\nabla f = \hat{\mathbf{r}}$, correct — the fastest way to increase your distance from the origin is to move directly away from it, at unit rate. For $f = \theta$ we get $\nabla f = \hat{\boldsymbol{\theta}}/r$, which correctly says that far from the origin you must travel a long way to change your bearing.

5 · The chain rule in several variables

Chapter 0.1 derived the chain rule by composing two linear approximations and observing that linear maps compose by multiplication. In one dimension "multiplication" meant multiplying two numbers. Here it means composing two linear maps, which by Chapter 0.4 means **multiplying two matrices**. The derivation is word-for-word the same.

Let $g : \mathbb{R}^n \to \mathbb{R}^p$ be differentiable at $\mathbf{a}$ and $f : \mathbb{R}^p \to \mathbb{R}^m$ differentiable at $\mathbf{b} = g(\mathbf{a})$. Nudge the input by $\mathbf{h}$. Then g moves by

$$\mathbf{k} \;\equiv\; g(\mathbf{a}+\mathbf{h}) - g(\mathbf{a}) \;=\; \mathrm{D}g(\mathbf{a})\,\mathbf{h} \;+\; o(|\mathbf{h}|). \tag{0.6.22}$$

Feed that displacement into f, using f's own linear approximation at $\mathbf{b}$:

$$\begin{aligned} f(\mathbf{b}+\mathbf{k}) &= f(\mathbf{b}) + \mathrm{D}f(\mathbf{b})\,\mathbf{k} + o(|\mathbf{k}|) \\ &= f(\mathbf{b}) + \mathrm{D}f(\mathbf{b})\Big[\mathrm{D}g(\mathbf{a})\,\mathbf{h} + o(|\mathbf{h}|)\Big] + o(|\mathbf{k}|) \\ &= f(\mathbf{b}) + \mathrm{D}f(\mathbf{b})\,\mathrm{D}g(\mathbf{a})\,\mathbf{h} + o(|\mathbf{h}|). \end{aligned} \tag{0.6.23}$$

Two small steps hide in that last line, and both are the boundedness lemma. First, $\mathrm{D}f(\mathbf{b})$ applied to something $o(|\mathbf{h}|)$ is again $o(|\mathbf{h}|)$. Second, $|\mathbf{k}| \le C\,|\mathbf{h}|$ for small $\mathbf{h}$ by (0.6.22), so anything that is $o(|\mathbf{k}|)$ is also $o(|\mathbf{h}|)$. Comparing the result with the definition (0.6.4) and invoking uniqueness (§2.1) to read off the coefficient:

$$\boxed{\mathrm{D}(f\circ g)(\mathbf{a}) \;=\; \mathrm{D}f\big(g(\mathbf{a})\big)\,\mathrm{D}g(\mathbf{a}).} \tag{0.6.24}$$

In matrices: an $(m \times p)$ Jacobian times a $(p \times n)$ Jacobian gives an $(m \times n)$ Jacobian. The inner dimensions match automatically, because the output space of g is the input space of f — the shapes cannot be wrong. This is one of the reasons to think of the derivative as a map rather than a table of numbers: the bookkeeping is enforced by the structure.

5.1 · The case you will use every day

Let $\mathbf{x}(t)$ be a curve in $\mathbb{R}^n$ and f a scalar field. Then $f \circ \mathbf{x}$ is a function of one variable, $\mathrm{D}\mathbf{x}(t)$ is the column $\dot{\mathbf{x}}$, and $\mathrm{D}f$ is the row of partials, so (0.6.24) reduces to a row times a column:

$$\frac{\mathrm{d}}{\mathrm{d}t} f\big(\mathbf{x}(t)\big) \;=\; \nabla f \cdot \dot{\mathbf{x}}(t) \;=\; \frac{\partial f}{\partial x^i}\,\frac{\mathrm{d}x^i}{\mathrm{d}t}, \tag{0.6.25}$$

which is (0.6.16), now obtained as a special case rather than from scratch. This is the workhorse. It is how you compute the rate of change of a temperature felt by a moving thermometer, of a potential felt by a moving charge, of a Lagrangian along a trajectory — Chapter 1.2 opens with it.

5.2 · Changing coordinates, and a convention appearing on its own

Now the case that runs all of tensor analysis. Suppose the x^i are themselves functions of new coordinates u^a — polar in terms of Cartesian, say, or one observer's coordinates in terms of another's. Then (0.6.24), written out entry by entry, is

$$\frac{\partial f}{\partial u^a} = \sum_i \frac{\partial f}{\partial x^i}\frac{\partial x^i}{\partial u^a} \equiv \frac{\partial f}{\partial x^i}\frac{\partial x^i}{\partial u^a}. \tag{0.6.26}$$

Three things about that equation deserve to be noticed now rather than in Chapter 2.4.

It is a matrix product. The object $\partial x^i/\partial u^a$ is the Jacobian of the coordinate change, and (0.6.26) says: to convert the components of $\mathrm{d}f$ from one coordinate system to another, hit them with that Jacobian. That single sentence, promoted to a definition, is what a tensor *is*.

The summation happens by itself. The index i appears twice on the right and is summed over; the index a appears once on each side and is not. That pattern is not a coincidence of this formula — it is what always happens when you compose linear maps, because matrix multiplication sums over the shared index. Chapter 2.4 will therefore drop the $\sum$ entirely and adopt the **Einstein summation convention**: a repeated index is summed. The convention is possible because the pattern is universal, and the pattern is universal because everything in sight is a composition of linear maps.

The index heights are already right. Look at where the repeated index i sits: upstairs in $\partial x^i/\partial u^a$ (in the numerator) and downstairs in $\partial f/\partial x^i$ (in the denominator). By the rule of §4 — differentiation lowers, coordinates raise — one is up and one is down, and the summed pair consists of one of each. This is not luck. It is (0.6.21) and (0.6.18) telling you that the only combinations that can be summed to give a coordinate-independent number are one-up-one-down. In Chapter 2.4 that becomes a rule you can use to check your algebra: if you have written down a formula with two indices at the same height summed against each other, you have made a mistake.

6 · Second derivatives, the Hessian, and what a critical point looks like

Differentiate a partial derivative again and you get a second partial, written $\partial_i\partial_j f$ or $\partial^2 f/\partial x^i \partial x^j$. There are n^2 of them, and the first question is whether the order matters.

6.1 · Clairaut's theorem, with the hypothesis said out loud

THEOREM (CLAIRAUT / SCHWARZ)

If the second partial derivatives of f exist in a neighbourhood of $\mathbf{a}$ and are **continuous at a**, then

$$\partial_i\partial_j f(\mathbf{a}) = \partial_j\partial_i f(\mathbf{a}) \qquad \text{for all } i, j.$$

The continuity hypothesis is real: the grind box below both proves the theorem and exhibits a function for which the two mixed partials at the origin are $+1$ and -1. But for every function you will meet in this book the hypothesis holds, so second partials commute and the matrix of them is symmetric. That matrix has a name:

$$H(\mathbf{a}) \;=\; \big[\partial_i \partial_j f(\mathbf{a})\big], \qquad H^{\mathsf{T}} = H. \tag{0.6.27}$$

It is the **Hessian**. Note what it is, structurally: ∇f is a map $\mathbb{R}^n \to \mathbb{R}^n$, and H is its Jacobian. The second derivative of a scalar field is the derivative of a vector field.

▼ Grind box — proof of Clairaut, and the counterexample when continuity fails

Proof. Work in the two relevant variables and call them x, y. Consider the *double difference* across a small rectangle of sides h, k:

$$\Delta \;=\; f(a+h, b+k) - f(a+h, b) - f(a, b+k) + f(a, b).$$

It is symmetric under swapping the roles of the two variables, and we will evaluate it in the two possible orders. First group it as a difference in x of the function $\varphi(x) = f(x, b+k) - f(x, b)$, so $\Delta = \varphi(a+h) - \varphi(a)$. By the mean value theorem (Chapter 0.2) there is $\theta \in (0,1)$ with $\Delta = h\,\varphi'(a+\theta h)$, i.e.

$$\Delta = h\Big[\partial_x f(a+\theta h,\, b+k) - \partial_x f(a+\theta h,\, b)\Big].$$

The bracket is a difference in y of the function $\partial_x f(a+\theta h, \cdot)$, so the mean value theorem applies again, giving $\theta' \in (0,1)$ with

$$\Delta = h\,k\;\partial_y \partial_x f\big(a+\theta h,\; b+\theta' k\big).$$

Now group Δ the other way, as a difference in y first. The identical argument gives $\sigma, \sigma' \in (0,1)$ with $\Delta = h\,k\;\partial_x \partial_y f(a+\sigma h, b+\sigma' k)$. Equate the two, divide by $hk \neq 0$, and let $(h,k) \to (0,0)$. Both evaluation points converge to (a,b), and by the assumed continuity of the two second partials *at* (a,b) both sides converge to their values there. Hence $\partial_y \partial_x f(a,b) = \partial_x \partial_y f(a,b)$. ∎

The counterexample. Continuity was used in the very last step, so removing it should break the theorem, and it does. Take

$$f(x,y) = \frac{xy\,(x^2 - y^2)}{x^2 + y^2} \;\text{ for } (x,y) \neq (0,0), \qquad f(0,0) = 0.$$

This one *is* continuous, and differentiable, and has second partials everywhere. Compute $\partial_x f$ in general and then restrict it to the y-axis: a short calculation gives $\partial_x f(0,y) = -y$. Restricting $\partial_y f$ to the x-axis gives $\partial_y f(x,0) = +x$. Therefore

$$\partial_y \partial_x f(0,0) = \frac{\mathrm{d}}{\mathrm{d}y}(-y) = -1, \qquad \partial_x \partial_y f(0,0) = \frac{\mathrm{d}}{\mathrm{d}x}(+x) = +1.$$

The mixed partials at the origin differ. What fails is precisely the hypothesis: away from the origin the second mixed partial is a bounded but direction-dependent function of θ — the same pathology as §1's counterexample, one derivative up — so it has no limit at the origin and is not continuous there. The theorem is not "obviously true"; it is true under a condition, and the condition is exactly what stops direction-dependence from surviving to the limit.

6.2 · Taylor to second order

We already have the machinery. Fix $\mathbf{a}$ and $\mathbf{h}$ and define the one-variable function $\varphi(t) = f(\mathbf{a} + t\mathbf{h})$, which traces f along the straight line through $\mathbf{a}$ in the direction $\mathbf{h}$. Differentiate it with the chain rule (0.6.25), noting that the "velocity" of the line is the constant vector $\mathbf{h}$:

$$\varphi'(t) = \nabla f(\mathbf{a} + t\mathbf{h}) \cdot \mathbf{h} = h^i\,\partial_i f(\mathbf{a} + t\mathbf{h}). \tag{0.6.28}$$

Differentiate again, applying the same rule to each $\partial_i f$:

$$\varphi''(t) = h^i h^j\,\partial_j\partial_i f(\mathbf{a} + t\mathbf{h}) = \mathbf{h}^\mathsf{T} H(\mathbf{a} + t\mathbf{h})\,\mathbf{h}. \tag{0.6.29}$$

Now apply the one-variable Taylor theorem of Chapter 0.3 to φ about $t = 0$ and evaluate at $t = 1$, which lands exactly on $f(\mathbf{a} + \mathbf{h})$:

$$\boxed{f(\mathbf{a} + \mathbf{h}) \;=\; f(\mathbf{a}) \;+\; \nabla f(\mathbf{a}) \cdot \mathbf{h} \;+\; \tfrac{1}{2}\,\mathbf{h}^\mathsf{T} H(\mathbf{a})\,\mathbf{h} \;+\; o(|\mathbf{h}|^2).} \tag{0.6.30}$$

Three terms: a constant, a linear form, a quadratic form. That is the general shape of a smooth function near a point, and it is why quadratic forms — Chapter 0.5's whole subject — are not an algebraic curiosity but the universal local model of everything.

6.3 · Critical points, classified by eigenvalues

A **critical point** is a point where $\nabla f = \mathbf{0}$: no direction, to first order, changes f. Every maximum, minimum and saddle is one, because if $\nabla f \neq \mathbf{0}$ you can increase f by stepping along it and decrease f by stepping the other way, so you are not at an extremum. At a critical point the linear term of (0.6.30) vanishes and

$$f(\mathbf{a} + \mathbf{h}) - f(\mathbf{a}) \;=\; \tfrac{1}{2}\,\mathbf{h}^\mathsf{T} H\,\mathbf{h} \;+\; o(|\mathbf{h}|^2). \tag{0.6.31}$$

So the local behaviour is decided by the sign of the quadratic form $\mathbf{h}^\mathsf{T} H\mathbf{h}$. And H is symmetric, which is exactly the hypothesis of the **spectral theorem** of Chapter 0.5: there is an orthonormal basis $\mathbf{e}_1, \ldots, \mathbf{e}_n$ of eigenvectors of H, with real eigenvalues $\mu_1, \ldots, \mu_n$. Expand the displacement in that basis, $\mathbf{h} = \sum_k c_k\mathbf{e}_k$, and the quadratic form collapses to a sum of squares:

$$\mathbf{h}^{\mathsf{T}} H \mathbf{h} \;=\; \sum_k \mu_k c_k^2, \qquad |\mathbf{h}|^2 = \sum_k c_k^2. \tag{0.6.32}$$

All cross terms are gone. The classification now reads itself off:

- **All $\mu_k > 0$:** then $\sum \mu_k c_k^2 \geq \mu_{\min} |\mathbf{h}|^2$ with $\mu_{\min} > 0$, so the quadratic term is positive and of order $|\mathbf{h}|^2$, while the remainder is $o(|\mathbf{h}|^2)$ and therefore eventually smaller. For small enough $|\mathbf{h}|$ the difference in (0.6.31) is positive in every direction: a strict local **minimum**.
- **All $\mu_k < 0$:** the same argument with signs reversed — a strict local **maximum**.
- **Mixed signs:** a **saddle**. Along an eigenvector with $\mu_k > 0$ the function rises; along one with $\mu_k < 0$ it falls. It is a maximum and a minimum at the same point, depending on which way you look.
- **Some $\mu_k = 0$:** the test is silent. The quadratic term vanishes in that direction and the $o(|\mathbf{h}|^2)$ remainder — which we have no control over — decides. You need higher order. Chapter 6.6's Higgs potential at the symmetric point is exactly such a case, and what happens in the flat direction is the entire physics.

The eigenvectors are not bystanders. Set $\mathbf{h} = t\mathbf{e}_k$ in (0.6.31): since $\mathbf{e}_k^{\mathsf{T}} H \mathbf{e}_k = \mu_k$,

$$f(\mathbf{a} + t\,\mathbf{e}_k) - f(\mathbf{a}) \;=\; \tfrac{1}{2}\mu_k t^2 + o(t^2). \tag{0.6.33}$$

Along the k-th eigenvector, f is a one-dimensional parabola whose second derivative is μ_k. **The eigenvalues of the Hessian are the curvatures of f along the principal directions, and the eigenvectors are those directions.** Near a minimum the level sets are ellipsoids: setting $\frac{1}{2}\sum \mu_k c_k^2 = \delta$ gives semi-axes $c_k = \sqrt{2\delta/\mu_k}$, so a *small* eigenvalue means a *long* axis — a shallow, extended valley in that direction. Hold onto the picture of a long thin valley; the next callout is about nothing else.

FAMILIAR GROUND

You already read Hessians for a living. When a model is fitted by maximum likelihood, the software finds $\hat{\boldsymbol{\theta}}$ where the gradient of the log-likelihood $\ell(\boldsymbol{\theta})$ vanishes, and then reports standard errors. Those standard errors are §6.3, exactly.

At the maximum, $H = \left[\partial_i \partial_j \ell(\hat{\boldsymbol{\theta}})\right]$ has all eigenvalues ≤ 0 — that is what "maximum" means. So $-H$ is positive semi-definite, and it gets its own name: the **observed Fisher information**, $J(\hat{\boldsymbol{\theta}}) = -H(\hat{\boldsymbol{\theta}})$. The estimated covariance matrix of the estimate is its inverse, and the reported standard errors are

$$\widehat{\mathrm{Cov}}(\hat{\boldsymbol{\theta}}) = J^{-1}, \qquad \mathrm{SE}_j = \sqrt{\left(J^{-1}\right)_{jj}}.$$

Read that as geometry and it says something you can feel: **the curvature of the log-likelihood surface at its peak is the precision of the estimate.** A sharp peak — large curvature, large J, small J^{-1} — is a tight confidence interval. A flat peak is a wide one. The data's ability to distinguish parameter values *is* the steepness with which the likelihood falls away from the best fit.

The one-parameter case, checkable by hand. For n exponentially distributed survival times with rate λ and no censoring, $\ell(\lambda) = n \ln \lambda - \lambda \sum_i t_i$. Setting $\ell' = 0$ gives $\hat{\lambda} = n / \sum t_i = 1/\bar{t}$, and $\ell''(\lambda) = -n/\lambda^2$, so $J = n/\hat{\lambda}^2$ and

$$\mathrm{SE}(\hat{\lambda}) = J^{-1/2} = \hat{\lambda}/\sqrt{n}.$$

The familiar $1/\sqrt{n}$ is the statement that doubling the sample doubles the curvature.

Observed versus expected — the distinction is not cosmetic. The *expected* Fisher information is $I(\boldsymbol{\theta}) = \mathbb{E}_{\boldsymbol{\theta}}[-H]$, an average of the curvature over all datasets the model could have produced. The *observed* information is $-H$ evaluated on the dataset you actually have, at the estimate you actually got. They agree in expectation, and both are consistent; in the exponential example above they happen to be identically equal, because $\ell'' = -n/\lambda^2$ contains no data. In general they differ, and Efron and Hinkley's argument — which has held up — is that the observed information is the better variance estimate for the sample in hand, because it conditions on what you actually saw rather than averaging over what you might have seen. Most software reports the observed version by default. It is worth knowing which one you are quoting.

The spectral reading, which is where Chapter 0.5 earns its place. Diagonalise $J = \sum_k \mu_k \mathbf{e}_k \mathbf{e}_k^{\mathsf{T}}$. Then $J^{-1} = \sum_k \mu_k^{-1} \mathbf{e}_k \mathbf{e}_k^{\mathsf{T}}$, and the variance of the linear combination $\mathbf{e}_k \cdot \hat{\boldsymbol{\theta}}$ is exactly $\mathbf{e}_k^{\mathsf{T}} J^{-1} \mathbf{e}_k = 1/\mu_k$. So: **the eigenvectors of the Hessian are the directions in parameter space along which the data constrain you best and worst, and the reciprocal eigenvalues are how badly.**

Two collinear covariates make this concrete. Fit $y = \beta_1 x_1 + \beta_2 x_2 + \varepsilon$ with Gaussian noise of known variance σ^2; then ℓ is $-\frac{1}{2\sigma^2}$ times a sum of squares, and $J = -H = X^{\mathsf{T}} X / \sigma^2$. Suppose $\sigma = 1$ and

$$J = \begin{pmatrix} 100 & 99 \\ 99 & 100 \end{pmatrix}, \qquad J^{-1} = \frac{1}{199} \begin{pmatrix} 100 & -99 \\ -99 & 100 \end{pmatrix}.$$

The eigenvalues are 199 (eigenvector $(1,1)/\sqrt{2}$) and 1 (eigenvector $(1,-1)/\sqrt{2}$). The *sum* $\beta_1+\beta_2$ is pinned down with variance $1/199$; the *difference* is known only with variance 1, two hundred times worse. That is the long thin valley of §6.3, and it is what "the two predictors are collinear" means geometrically: the likelihood has a nearly flat direction. The reported standard errors are $\sqrt{100/199}=0.709$ each — seven times larger than the naive $1/\sqrt{100}=0.1$ you would get by inverting the diagonal instead of taking the diagonal of the inverse. That factor of seven *is* the correlation, here $-99/100=-0.99$. Inverting first and taking the diagonal second is not a convention; it is the only way the off-diagonal curvature can reach the answer.

7 · Lagrange multipliers

Now the constrained problem: extremise $f(\mathbf{x})$ subject to $g(\mathbf{x})=c$. The recipe is famous and usually presented as one; we will derive it instead, in three sentences of geometry, and then extract something from it that the recipe hides.

Let $\mathbf{a}$ be a point on the constraint surface $S=\{g=c\}$ at which f is extremal *among points of* S, and assume $\nabla g(\mathbf{a})\neq\mathbf{0}$. Two facts, both already proved.

Which directions are tangent to S. If $\mathbf{x}(t)$ is a curve lying in S then $g(\mathbf{x}(t))=c$, so by (0.6.16), $\nabla g\cdot\dot{\mathbf{x}}=0$: every tangent vector to S is orthogonal to ∇g. The converse — that every vector orthogonal to ∇g is the velocity of some curve in S — is the implicit function theorem, which we quote; it is guaranteed precisely by the assumption $\nabla g\neq\mathbf{0}$, which says the constraint really does cut down the dimension by one near $\mathbf{a}$. So the tangent space to S at $\mathbf{a}$ is exactly $(\operatorname{span}\nabla g)^{\perp}$.

What extremality says. If $\mathbf{x}(t)$ is a curve in S through $\mathbf{a}$, then $t\mapsto f(\mathbf{x}(t))$ has an extremum at that point, so its derivative vanishes: $\nabla f\cdot\dot{\mathbf{x}}=0$. Moving along the constraint surface cannot change f to first order — that is what being extremal *on* S means. Hence $\nabla f(\mathbf{a})$ is orthogonal to every tangent vector of S.

Put them together, using Chapter 0.5's projection. Write $\mathbf{n}=\nabla g(\mathbf{a})$ and split ∇f into its component along $\mathbf{n}$ and whatever is left:

$$\nabla f \;=\; \lambda\,\mathbf{n} \;+\; \mathbf{w}, \qquad \lambda \;\equiv\; \frac{\nabla f\cdot\mathbf{n}}{\mathbf{n}\cdot\mathbf{n}}, \qquad \mathbf{w}\;\equiv\;\nabla f-\lambda\mathbf{n}. \tag{0.6.34}$$

The residual is orthogonal to $\mathbf{n}$ by construction — $\mathbf{w}\cdot\mathbf{n}=\nabla f\cdot\mathbf{n}-\lambda\,|\mathbf{n}|^2=0$ — so by the first fact $\mathbf{w}$ is a tangent vector to S. But the second fact says ∇f is orthogonal to *every* tangent vector, so in particular to $\mathbf{w}$ itself:

$$0 \;=\; \nabla f\cdot\mathbf{w} \;=\; \big(\lambda\mathbf{n}+\mathbf{w}\big)\cdot\mathbf{w} \;=\; \lambda\,\underbrace{\mathbf{n}\cdot\mathbf{w}}_{0}+|\mathbf{w}|^2 \;=\; |\mathbf{w}|^2. \tag{0.6.35}$$

A vector of zero length is the zero vector, so $\mathbf{w}=\mathbf{0}$ and

$$\boxed{\nabla f(\mathbf{a}) \;=\; \lambda\,\nabla g(\mathbf{a}), \qquad g(\mathbf{a})=c.} \tag{0.6.36}$$

Notice that the argument even handed us the multiplier explicitly, $\lambda = \nabla f \cdot \nabla g / |\nabla g|^2$, which is worth remembering when you want to check a solution rather than find one. The geometric statement is the one to remember: **at a constrained extremum the level set of f is tangent to the constraint surface**, since both have the same normal direction. If they crossed instead of touching, you could slide along the constraint from one side of the f-contour to the other and change f — so you were not at an extremum. Press the button in the §3 figure and this is exactly what you are looking at.

The mechanical version everyone quotes follows immediately. Define

$$\mathcal{L}(\mathbf{x}, \lambda) \;=\; f(\mathbf{x}) \;-\; \lambda\big(g(\mathbf{x}) - c\big) \tag{0.6.37}$$

and look for its *unconstrained* critical points. Setting $\partial\mathcal{L}/\partial x^i = 0$ gives $\nabla f = \lambda \nabla g$; setting $\partial\mathcal{L}/\partial\lambda = 0$ gives $g = c$. The constraint has been converted from a restriction on the search space into one more equation to solve, at the price of one more unknown. That trade — restrict the space, or enlarge it and add a multiplier — is one of the deepest moves in theoretical physics, and it returns in Chapter 1.2 as the way to handle a bead on a wire, and again, much later, as the way gauge symmetry is imposed.

7.1 · The multiplier means something

λ looks like a bookkeeping device you introduce and then discard. It is not. Let $\mathbf{x}^*(c)$ be the constrained optimum as a function of the constraint level c, and let $f^*(c) = f(\mathbf{x}^*(c))$ be the optimal value. Assume $\mathbf{x}^*$ varies differentiably with c (it does, under the same non-degeneracy that made the multiplier exist). Differentiate f^* using the chain rule (0.6.25):

$$\frac{\mathrm{d}f^*}{\mathrm{d}c} \;=\; \nabla f \cdot \frac{\mathrm{d}\mathbf{x}^*}{\mathrm{d}c} \;=\; \lambda\, \nabla g \cdot \frac{\mathrm{d}\mathbf{x}^*}{\mathrm{d}c}, \tag{0.6.38}$$

where the second step used (0.6.36). Now differentiate the constraint itself, $g(\mathbf{x}^*(c)) = c$, with respect to c — the same chain rule, applied to an identity that holds for all c:

$$\nabla g \cdot \frac{\mathrm{d}\mathbf{x}^*}{\mathrm{d}c} \;=\; 1. \tag{0.6.39}$$

Substituting (0.6.39) into (0.6.38):

$$\boxed{\frac{\mathrm{d}f^*}{\mathrm{d}c} \;=\; \lambda.} \tag{0.6.40}$$

The multiplier is a **sensitivity**: it is the rate at which the best achievable value improves when you relax the constraint by one unit. Economists call it a shadow price; in mechanics it is the constraint force; in thermodynamics, as Worked Example 1 will show, it is the temperature. A large λ says the constraint is expensive and worth negotiating; $\lambda = 0$ says the constraint is not binding at all.

A concrete check, which is also the figure's setup. Minimise $f = x^2 + 2y^2$ subject to $x + y = c$. Then (0.6.36) reads $(2x, 4y) = \lambda(1, 1)$, so $x = \lambda/2$ and $y = \lambda/4$; the constraint gives $3\lambda/4 = c$, hence $\lambda = 4c/3$, $\mathbf{x}^* = (2c/3,\ c/3)$

and

$$f^*(c) = \frac{4c^2}{9} + \frac{2c^2}{9} = \frac{2c^2}{3}, \qquad \frac{\mathrm{d}f^*}{\mathrm{d}c} = \frac{4c}{3} = \lambda. \checkmark \tag{0.6.41}$$

At $c = 3$: the optimum is $(2, 1)$, $f^* = 6$, and $\lambda = 4$ — which is the number printed in the figure when the constraint is showing.

▼ Grind box — when Lagrange fails, and why the fine print is the interesting part

The derivation used $\nabla g(\mathbf{a}) \neq \mathbf{0}$ twice: to know the tangent space is a genuine hyperplane, and to invoke the implicit function theorem. Drop it and the method can simply be wrong. Minimise $f(x, y) = x$ subject to

$$g(x, y) = x^3 - y^2 = 0.$$

The constraint set is the curve $y^2 = x^3$, a cusp at the origin, defined only for $x \geq 0$. The minimum of x on it is obviously at the origin. But there $\nabla g = (3x^2, -2y) = (0, 0)$, while $\nabla f = (1, 0)$ everywhere, and no λ satisfies $(1, 0) = \lambda(0, 0)$. The method returns nothing, and the answer it failed to find was sitting at the origin all along.

The condition $\nabla g \neq \mathbf{0}$ is called a *constraint qualification*, and the moral is that the multiplier exists because the constraint surface is smooth and non-degenerate there — not because algebra says so. Where constraints are degenerate, the multiplier framework needs repair. That is not an exotic case: in gauge theories the constraints are degenerate *by design*, because a gauge symmetry means several different field configurations describe the same physics, and handling that properly is what Dirac's constrained-Hamiltonian machinery and the Faddeev–Popov procedure exist to do. Every one of those is a descendant of (0.6.36) plus an honest accounting of when it breaks.

Several constraints. With $g_1 = c_1, \ldots, g_r = c_r$ the argument is identical word for word, with $\mathbf{n}$ replaced by the subspace $W = \operatorname{span}\{\nabla g_1, \ldots, \nabla g_r\}$ and the one-line split (0.6.34) replaced by Chapter 0.5's orthogonal projection onto W. Tangent vectors are those orthogonal to every ∇g_α; the residual $\mathbf{w} = \nabla f - P\nabla f$ is one of them; extremality kills it; and what survives is

$$\nabla f = \sum_{\alpha=1}^{r} \lambda_\alpha \nabla g_\alpha,$$

with the qualification now reading "the ∇g_α are linearly independent at $\mathbf{a}$" — which is Chapter 0.4's notion of independence doing real work. Worked Example 1 uses two constraints and therefore two multipliers, one of which turns out to be the temperature.

8 · Change of variables and the Jacobian determinant

The last consequence, and the one that pays a debt. Let $\Phi : U \to \mathbb{R}^n$ be a differentiable, injective change of coordinates with $\det \mathrm{D}\Phi \neq 0$. How does an integral transform?

The answer is forced by two things we already have. *First*, §2: near any point $\mathbf{u}$, the map Φ **is** a linear map, up to an error that dies faster than the displacement,

$$\Phi(\mathbf{u} + \mathbf{h}) = \Phi(\mathbf{u}) + \mathrm{D}\Phi(\mathbf{u})\,\mathbf{h} + o(|\mathbf{h}|). \tag{0.6.42}$$

Second, Chapter 0.4: a linear map A takes a region of volume V to a region of volume $|\det A|\ V$ — that is what the determinant *is*, the signed factor by which a linear map scales volume, and the absolute value is there because volume does not care about orientation.

Put them together. Chop U into tiny cubes of side ϵ. The cube at $\mathbf{u}$ is a set of displacements $\mathbf{h}$ of size at most $\epsilon\sqrt{n}$, so by (0.6.42) its image under Φ is the image under the linear map $\mathrm{D}\Phi(\mathbf{u})$ — a parallelepiped — translated to $\Phi(\mathbf{u})$, with a distortion that is small *compared to* ϵ. Hence

$$\mathrm{vol}\big(\Phi(\mathrm{cube})\big) \;=\; |\det \mathrm{D}\Phi(\mathbf{u})|\ \epsilon^n\,\big(1 + o(1)\big). \tag{0.6.43}$$

Summing $f(\Phi(\mathbf{u}))$ times that over all cubes and taking $\epsilon \to 0$ turns the Riemann sums of Chapter 0.2 on one side into the Riemann sums on the other, giving

$$\boxed{\int_{\Phi(U)} f(\mathbf{x})\,\mathrm{d}^n x \;=\; \int_U f\big(\Phi(\mathbf{u})\big)\ |\det \mathrm{D}\Phi(\mathbf{u})|\ \mathrm{d}^n u.} \tag{0.6.44}$$

In one dimension $\det \mathrm{D}\Phi = \Phi'$ and this is the substitution rule of Chapter 0.2, with the absolute value doing the job that reversing the limits of integration does there. The factor $|\det \mathrm{D}\Phi|$ is called the **Jacobian determinant** and it is the local volume-magnification of the coordinate change: nothing more mysterious than that.

▼ Grind box — what that argument does and doesn't establish

The chop-and-sum argument is the right picture and it is how every physicist thinks about the Jacobian, but it is not a proof, and it is worth being clear about where the gap is.

Two things were glossed. **(i) Uniformity.** The $o(1)$ in (0.6.43) must shrink at a rate that does not depend on which cube you are looking at, or summing a large number of small errors need not give a small total. This is delivered by continuity of $\mathrm{D}\Phi$ on a compact region, which is the usual hypothesis. **(ii) The boundary.** Cubes that straddle the edge of U are only partly inside, and one must show their total contribution vanishes — true, because the boundary of a reasonable region has n-dimensional volume zero, but "reasonable" needs defining, and doing it properly is where measure theory enters and where the theorem is genuinely a theorem.

What the argument *does* establish, and what you should take from it, is the mechanism: the $|\det|$ appears because differentiability means local linearity, and because determinants are what linear maps do to volume. Change either fact and the formula changes with it.

8.1 · Polar coordinates — paying Chapter 0.2's debt

In Chapter 0.2, deriving the Gaussian integral, we needed the polar area element and obtained it geometrically, by differencing two circular sectors. It was an honest argument, but it was special to the plane and to circles, and the chapter explicitly said that Chapter 0.6 would produce the same factor mechanically. Here is that promise kept.

Take $\Phi(r,\theta) = (r\cos\theta,\ r\sin\theta)$. Its Jacobian is a 2×2 matrix of partials, computed by the rules of §1 — differentiate each output with respect to each input:

$$\mathrm{D}\Phi = \begin{pmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{pmatrix} = \begin{pmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{pmatrix}. \tag{0.6.45}$$

Its determinant, by the 2×2 rule $ad - bc$ of Chapter 0.4, is

$$\det \mathrm{D}\Phi = r\cos^2\theta + r\sin^2\theta = r, \tag{0.6.46}$$

and therefore, since $r \geq 0$,

$$\mathrm{d}x\,\mathrm{d}y \;=\; r\,\mathrm{d}r\,\mathrm{d}\theta. \tag{0.6.47}$$

That is the factor Chapter 0.2 used on faith, now derived from the definition of the determinant. The debt is paid, and the derivation cost three lines instead of a picture. Note that it also *explains* the geometric argument: the columns of (0.6.45) are the displacement produced by a unit change in r (length 1, pointing radially) and by a unit change in θ (length r, pointing tangentially). They are perpendicular, so the parallelogram they span has area $1\times r = r$. The determinant computed the area of the little polar cell, which is precisely what the sector argument computed by hand.

8.2 · Spherical coordinates

Same procedure in three dimensions, with $x = r\sin\theta\cos\phi, y = r\sin\theta\sin\phi, z = r\cos\theta$ (θ measured from the z-axis, as physics always does):

$$\mathrm{D}\Phi = \begin{pmatrix} \sin\theta\cos\phi & r\cos\theta\cos\phi & -r\sin\theta\sin\phi \\ \sin\theta\sin\phi & r\cos\theta\sin\phi & r\sin\theta\cos\phi \\ \cos\theta & -r\sin\theta & 0 \end{pmatrix}. \tag{0.6.48}$$

Expand along the bottom row, which has a zero in it and so costs only two cofactors. The $\cos\theta$ entry multiplies

$$\det \begin{pmatrix} r\cos\theta\cos\phi & -r\sin\theta\sin\phi \\ r\cos\theta\sin\phi & r\sin\theta\cos\phi \end{pmatrix} = r^2 \sin\theta\cos\theta, \tag{0.6.49}$$

after using $\cos^2\phi + \sin^2\phi = 1$; and the $-r\sin\theta$ entry, with the alternating sign $(-1)^{3+2} = -1$, multiplies $-\left(r\sin^2\theta\right)$. Adding the two contributions:

$$\det \mathrm{D}\Phi = r^2\sin\theta\cos^2\theta + r^2\sin^3\theta = r^2\sin\theta, \tag{0.6.50}$$

using $\cos^2\theta + \sin^2\theta = 1$ once more. Hence

$$\mathrm{d}V = r^2\sin\theta\ \mathrm{d}r\ \mathrm{d}\theta\ \mathrm{d}\phi, \tag{0.6.51}$$

positive for $\theta \in (0, \pi)$, so the absolute value is automatic. Problem 1 uses this to compute the volume of a sphere, which is the cheapest possible check that the factor is right.

8.3 · The same factor, in general relativity

One forward glance, because the connection is exact rather than analogical. In curved spacetime the invariant volume element is written $\sqrt{-g}\ \mathrm{d}^4x$, where $g = \det g_{\mu\nu}$ is the determinant of the metric (negative because of the Lorentzian signature, hence the minus sign under the root). That $\sqrt{-g}$ is (0.6.44) wearing different clothes.

You can see it without any relativity at all, using the flat-space metrics of §4. In plane polar coordinates $g_{ij} = \mathrm{diag}(1, r^2)$, so

$$\sqrt{\det g_{ij}} = \sqrt{1\cdot r^2} = r, \tag{0.6.52}$$

which is exactly (0.6.47). In spherical coordinates the metric is $\mathrm{d}s^2 = \mathrm{d}r^2 + r^2\mathrm{d}\theta^2 + r^2\sin^2\theta\ \mathrm{d}\phi^2$, so $g_{ij} = \mathrm{diag}(1,\ r^2,\ r^2\sin^2\theta)$ and

$$\sqrt{\det g_{ij}} = \sqrt{r^4\sin^2\theta} = r^2\sin\theta, \tag{0.6.53}$$

exactly (0.6.51). The reason is structural: Chapter 3.3 will show that under a coordinate change the metric picks up two factors of the Jacobian matrix, so its determinant picks up $(\det J)^2$ and $\sqrt{|g|}$ picks up $|\det J|$ — precisely cancelling the $|\det J|^{-1}$ that d^nx acquires. The combination $\sqrt{|g|}\ \mathrm{d}^nx$ is therefore the same in every coordinate system, which is what "invariant volume element" means. The Einstein–Hilbert action of Chapter 3.6 carries that factor for this reason and no other.

The same $|\det|$ turns up every time you change variables in a path integral (Chapter 5.5), where it is no longer a nuisance factor but the whole story: the Faddeev–Popov ghosts of non-abelian gauge theory are the Jacobian determinant of a change of variables, promoted to a field.

9 · Worked examples

WORKED EXAMPLE 1 — THE BOLTZMANN DISTRIBUTION, FROM TWO CONSTRAINTS

A system can be in states $1, \dots, N$ with energies E_i. All you know is that the probabilities sum to one and that the mean energy is $\bar{E}$. What distribution should you assign?

The principle — due to Gibbs, sharpened by Jaynes — is to choose the distribution that maximises the entropy $S = -\sum_i p_i \ln p_i$, i.e. the one that is maximally noncommittal about everything you were not told. So: maximise S subject to

$$g_1 = \sum_i p_i = 1, \qquad g_2 = \sum_i p_i E_i = \bar{E}.$$

Two constraints, two multipliers. Form the Lagrangian of the grind box in §7, calling the multipliers α and β:

$$\mathcal{L} = -\sum_i p_i \ln p_i \; - \; \alpha\Big(\sum_i p_i - 1\Big) \; - \; \beta\Big(\sum_i p_i E_i - \bar{E}\Big).$$

The variables are the N numbers p_i, so take N partial derivatives. Using $\frac{\partial}{\partial p_i}\big(p_i \ln p_i\big) = \ln p_i + 1$:

$$\frac{\partial \mathcal{L}}{\partial p_i} = -\ln p_i - 1 - \alpha - \beta E_i = 0 \qquad \Longrightarrow \qquad p_i = \mathrm{e}^{-1-\alpha}\,\mathrm{e}^{-\beta E_i}.$$

The first factor is the same for every state, so it is fixed by normalisation. Imposing $\sum_i p_i = 1$ gives $\mathrm{e}^{-1-\alpha} = 1/Z$ with

$$\boxed{p_i = \frac{\mathrm{e}^{-\beta E_i}}{Z}, \qquad Z(\beta) = \sum_i \mathrm{e}^{-\beta E_i}.}$$

That is the Boltzmann distribution, and Z — which arrived as nothing more grand than a normalisation constant — is the **partition function**, from which all of equilibrium thermodynamics is extracted by differentiation. For instance $\bar{E} = -\partial_\beta \ln Z$, which is the second constraint restated and is what implicitly determines β.

Is it a maximum? §6 answers this without any extra work. The Hessian of S with respect to the p_i is $\partial_i \partial_j S = -\delta_{ij}/p_i$, a diagonal matrix with all entries negative, so all its eigenvalues are negative and the critical point is a genuine maximum. (Being diagonal, the spectral theorem has nothing to do — the eigenvalues are the diagonal entries. It is still §6.3 doing the work.)

What β is. It arrived as a multiplier on the energy constraint. By (0.6.40), a multiplier is the sensitivity of the optimum to its constraint level, so here

$$\beta = \frac{\mathrm{d}S^*}{\mathrm{d}\bar{E}},$$

the rate at which the maximum attainable entropy rises when you feed the system more energy. Now compare with the thermodynamic definition of temperature, $1/T = \frac{\partial S_{\text{thermo}}}{\partial U}$, and note that thermodynamic entropy is k_B times the information entropy used above. Then $1/T = k_B \beta$, so

$$\beta = \frac{1}{k_B T}.$$

This is the honest route to that identification: β is not *defined* as $1/k_B T$ and then justified, it *emerges* as a Lagrange multiplier and is *identified* with inverse temperature because §7.1 says multipliers are sensitivities and thermodynamics says $\partial S/\partial U$ is one over the temperature. Temperature is a shadow price.

Two remarks worth carrying. First, that calculation is most of statistical mechanics: everything else is evaluating Z for particular energy spectra and differentiating it. Second, look at the *shape* of the answer — a probability proportional to $\mathrm{e}^{-\beta(\text{something})}$, normalised by a sum over all possibilities. In Chapter 5.8 the path integral assigns to each history a weight $\mathrm{e}^{\mathrm{i}S/\hbar}$ normalised by a sum over all histories. The structures are the same down to the last symbol, with $\mathrm{i}S/\hbar$ in place of $-\beta E$; and the substitution $t \to -\mathrm{i}\tau$ that converts one into the other is the reason a quantum field theory at finite temperature and a statistical mechanics problem in one extra dimension are the same computation.

WORKED EXAMPLE 2 — CLASSIFYING EVERY CRITICAL POINT OF A DOUBLE WELL

Find and classify all critical points of $f(x,y) = x^4 + y^4 - 4xy$.

Locate them. Set the gradient to zero:

$$\partial_x f = 4x^3 - 4y = 0, \qquad \partial_y f = 4y^3 - 4x = 0 \qquad \Longrightarrow \qquad y = x^3, \;\; x = y^3.$$

Substituting the first into the second, $x = x^9$, so $x\left(x^8 - 1\right) = 0$, giving $x = 0$ or $x = \pm 1$ (over the reals). The critical points are

$$(0,0), \qquad (1,1), \qquad (-1,-1),$$

with $f(0,0) = 0$ and $f(\pm 1, \pm 1) = 1 + 1 - 4 = -2$.

The Hessian. Differentiating again,

$$H(x,y) = \begin{pmatrix} 12x^2 & -4 \\ -4 & 12y^2 \end{pmatrix},$$

symmetric as Clairaut promised — and note we may as well have checked directly: $\partial_y \partial_x f = \partial_y(4x^3 - 4y) = -4$ and $\partial_x \partial_y f = \partial_x(4y^3 - 4x) = -4$. ✓

At the origin. $H = \begin{pmatrix} 0 & -4 \\ -4 & 0 \end{pmatrix}$. Its eigenvalues solve $\det(H - \mu I) = \mu^2 - 16 = 0$, so $\mu = \pm 4$, with eigenvectors found by inspection: $H(1,-1)^{\mathsf{T}} = (4,-4)^{\mathsf{T}}$ and $H(1,1)^{\mathsf{T}} = (-4,-4)^{\mathsf{T}}$, so

$$\mu_+ = +4 \;\text{ along }\; \tfrac{1}{\sqrt{2}}(1,-1), \qquad \mu_- = -4 \;\text{ along }\; \tfrac{1}{\sqrt{2}}(1,1).$$

Mixed signs: a **saddle**. And (0.6.33) says exactly what it should look like. Check it against the function directly. Along the displacement (t,t),

$$f(t,t) = t^4 + t^4 - 4t \cdot t = 2t^4 - 4t^2,$$

which falls for small t; along $(t,-t)$, $f = 2t^4 + 4t^2$, which rises. And the quadratic terms match the prediction: the displacement (t,t) has component $c = \mathbf{h} \cdot \frac{1}{\sqrt{2}}(1,1) = t\sqrt{2}$ along the unit eigenvector, so (0.6.31) predicts $\frac{1}{2}\mu_- c^2 = \frac{1}{2}(-4)(2t^2) = -4t^2$ — exactly the $-4t^2$ above. ✓ Likewise $\frac{1}{2}(+4)(2t^2) = +4t^2$ in the other direction.

At $(1,1)$. $H = \begin{pmatrix} 12 & -4 \\ -4 & 12 \end{pmatrix}$. Same eigenvectors — any matrix of the form $aI + bJ$ with $J = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ has eigenvectors $(1,1)$ and $(1,-1)$ — with eigenvalues

$$\mu = 12 \mp 4: \qquad \mu_1 = 8 \;\text{ along }\; \tfrac{1}{\sqrt{2}}(1,1), \qquad \mu_2 = 16 \;\text{ along }\; \tfrac{1}{\sqrt{2}}(1,-1).$$

Both positive: a **local minimum**. The principal curvatures are 8 and 16, so the valley floor is twice as stiff across the diagonal as along it, and the elliptical contours around the minimum have semi-axis ratio $\sqrt{16/8} = \sqrt{2}$ — longer along $(1,1)$, which is the soft direction. By the symmetry $f(-x,-y) = f(x,y)$ the point $(-1,-1)$ is an identical minimum.

What you have just drawn. Two degenerate minima of equal depth, related by a symmetry, with a saddle sitting at the symmetric point between them. That is the double well, and it is the entire mechanism of spontaneous symmetry breaking: the potential respects the symmetry $(x, y) \to (-x, -y)$, but the system must sit in one minimum or the other and therefore does not. The soft direction along $(1, 1)$ at the minimum — the small eigenvalue — is the direction along which the system is easiest to excite, which in Chapter 0.8 becomes the lowest normal mode, and in Chapter 6.6, when the symmetry is continuous rather than discrete so that the soft direction becomes exactly flat, becomes a massless Goldstone boson. The eigenvalues of a Hessian are, quite literally, masses squared.

10 · Your turn

PROBLEM 1 — THE SPHERICAL VOLUME ELEMENT, AND A CHECK

Verify the Jacobian determinant $r^2 \sin\theta$ for spherical coordinates by expanding (0.6.48) along its *first* column instead of its last row (you should of course get the same answer). Then use (0.6.51) to compute the volume of a ball of radius R, and the surface area of a sphere of radius R.

− Solution

The determinant. Expanding along the first column, with entries $\sin\theta\cos\phi$, $\sin\theta\sin\phi$, $\cos\theta$ and signs $+, -, +$:

$$\begin{aligned} \det = \;& \sin\theta\cos\phi\,\left(0 + r^2\sin^2\theta\cos\phi\right) \\ & - \sin\theta\sin\phi\,\left(0 - r^2\sin^2\theta\sin\phi\right) \\ & + \cos\theta\,\left(r^2\sin\theta\cos\theta\cos^2\phi + r^2\sin\theta\cos\theta\sin^2\phi\right). \end{aligned}$$

The first two lines give $r^2\sin^3\theta(\cos^2\phi + \sin^2\phi) = r^2\sin^3\theta$; the third gives $r^2\sin\theta\cos^2\theta$. Adding, $\det = r^2\sin\theta(\sin^2\theta + \cos^2\theta) = r^2\sin\theta$. ✓ Same answer, as it must be — the determinant does not know which row you expanded along.

Volume. The integrand factorises completely, so the triple integral is a product:

$$V = \int_0^R\int_0^\pi\int_0^{2\pi} r^2\sin\theta\;\mathrm{d}\phi\,\mathrm{d}\theta\,\mathrm{d}r = \left(\int_0^R r^2\mathrm{d}r\right)\left(\int_0^\pi \sin\theta\,\mathrm{d}\theta\right)\left(\int_0^{2\pi}\mathrm{d}\phi\right).$$

These are $R^3/3$, $\big[-\cos\theta\big]_0^\pi = 2$, and 2π. So

$$V = \frac{R^3}{3}\cdot 2\cdot 2\pi = \frac{4}{3}\pi R^3.\ ✓$$

Area. Fix $r = R$; the surface element is what remains after dropping $\mathrm{d}r$ and setting $r = R$, namely $\mathrm{d}A = R^2\sin\theta\,\mathrm{d}\theta\,\mathrm{d}\phi$, so $A = R^2\cdot 2\cdot 2\pi = 4\pi R^2$. Note that $\frac{\mathrm{d}V}{\mathrm{d}R} = 4\pi R^2 = A$, which is not a coincidence: growing the ball by $\mathrm{d}R$ adds a shell of area A and thickness $\mathrm{d}R$. That is the fundamental theorem of Chapter 0.2 in three dimensions, and Chapter 0.7 turns the observation into the divergence theorem.

The $\sin\theta$ is doing something physical, incidentally: it says that a patch of $\mathrm{d}\theta\times\mathrm{d}\phi$ near the pole covers less area than the same patch at the equator, because the circles of constant θ shrink to points there. Any map projection you have ever complained about is a fight with this factor.

PROBLEM 2 — THE COULOMB FIELD AS A GRADIENT

Let $\phi(\mathbf{r}) = 1/r$ where $r = |\mathbf{r}| = \sqrt{x^2+y^2+z^2}$, defined for $\mathbf{r} \neq \mathbf{0}$. (a) Compute $\nabla\phi$. (b) Verify that it is perpendicular to the level sets of ϕ. (c) At the point $(1,2,2)$, give the direction of steepest ascent of ϕ and the rate of increase in that direction. (d) Interpret $-\nabla\phi$.

− Solution

(a) First $\frac{\partial r}{\partial x}$. From $r^2 = x^2+y^2+z^2$, differentiate both sides with respect to x: $2r\,\frac{\partial r}{\partial x} = 2x$, so $\frac{\partial r}{\partial x} = x/r$, and likewise for y, z. Compactly, $\nabla r = \mathbf{r}/r = \hat{\mathbf{r}}$ — the direction of fastest increase of distance from the origin is straight outward, at unit rate, which is reassuring. Now the chain rule of §5.1 with $\phi = r^{-1}$:

$$\nabla\phi = \frac{\mathrm{d}\phi}{\mathrm{d}r}\,\nabla r = -\frac{1}{r^2}\,\hat{\mathbf{r}} = -\frac{\mathbf{r}}{r^3}.$$

(b) The level sets $1/r = c$ are the spheres $r = 1/c$. A curve on such a sphere has $|\mathbf{x}(t)|$ constant, so differentiating $\mathbf{x}\cdot\mathbf{x} = \text{const}$ gives $2\,\mathbf{x}\cdot\dot{\mathbf{x}} = 0$: tangent vectors to a sphere are perpendicular to the radius. And $\nabla\phi$ is parallel to the radius. Hence $\nabla\phi\cdot\dot{\mathbf{x}} = 0$. ✓ — which is (0.6.17) confirmed in a case where you can see both objects.

(c) At $(1,2,2)$, $r = \sqrt{1+4+4} = 3$, so

$$\nabla\phi = -\frac{(1,2,2)}{27}, \qquad |\nabla\phi| = \frac{3}{27} = \frac{1}{9} = \frac{1}{r^2}.$$

By §3.1 the steepest-ascent direction is $\nabla\phi/\left|\nabla\phi\right| = -\frac{1}{3}(1,2,2)$ — a unit vector pointing directly *at* the origin — and the rate is $|\nabla\phi| = 1/9$. Obvious in hindsight: $1/r$ increases fastest when you head straight for the singularity.

(d) $-\nabla\phi = \mathbf{r}/r^3 = \hat{\mathbf{r}}/r^2$. That is an inverse-square field pointing radially outward — the electric field of a positive point charge (in units with $q/4\pi\epsilon_0 = 1$), or minus the gravitational field of a point mass. The relation $\mathbf{E} = -\nabla\phi$ is the definition of the potential, and the minus sign is the statement that *things fall down the gradient*: forces point in the direction of steepest *descent* of the potential energy, which by §3.1 is the direction $-\nabla\phi/\left|\nabla\phi\right|$. Chapter 0.7 takes this field and computes its flux; Chapter 1.2 takes the same ϕ and derives the orbit.

PROBLEM 3 — MAXIMUM ENTROPY WITH NOTHING TO GO ON

Repeat Worked Example 1 with only the normalisation constraint $\sum_i p_i = 1$: maximise $S = -\sum_i p_i \ln p_i$ over N states. Show that the answer is the uniform distribution and find the maximum entropy. Then obtain the same result as a limit of the Boltzmann distribution, and say what physical situation that limit describes.

− Solution

One constraint, one multiplier:

$$\mathcal{L} = -\sum_i p_i \ln p_i - \alpha\Big(\sum_i p_i - 1\Big), \qquad \frac{\partial \mathcal{L}}{\partial p_i} = -\ln p_i - 1 - \alpha = 0.$$

So $p_i = \mathrm{e}^{-1-\alpha}$ — *the same value for every i*, because the right-hand side carries no i at all. That single observation is the whole result. Normalising, $N\mathrm{e}^{-1-\alpha} = 1$, so

$$p_i = \frac{1}{N}, \qquad S_{\max} = -\sum_{i=1}^{N} \frac{1}{N} \ln \frac{1}{N} = \ln N.$$

The Hessian is again $\operatorname{diag}(-1/p_i) = \operatorname{diag}(-N)$, negative definite, so it is a maximum (§6.3). "Maximum entropy = uniform" is thus a one-line consequence of §7, not a separate principle.

As a limit. Boltzmann gives $p_i = \mathrm{e}^{-\beta E_i}/Z$. As $\beta \to 0$ every exponential tends to 1 and $Z \to N$, so $p_i \to 1/N$. Since $\beta = 1/k_B T$, the limit $\beta \to 0$ is $T \to \infty$: **at infinite temperature every state is equally likely**, because the energy constraint has stopped costing anything. Consistently, (0.6.40) reads $\beta = \frac{\mathrm{d}S^*}{\mathrm{d}\bar{E}}$, and at the unconstrained maximum S^* is stationary in $\bar{E}$, so $\beta = 0$. The two statements are the same statement.

Why this matters beyond the algebra. The result $p_i = 1/N$ is the "equal a priori probabilities" postulate of the microcanonical ensemble — the assumption usually placed at the very foundation of statistical mechanics and defended with hand-waving about ergodicity. Here it is a *theorem*: given only that the probabilities sum to one, the least presumptuous distribution is the flat one. And $S_{\max} = \ln N$ is Boltzmann's $S = k_B \ln W$ carved on his headstone, with $W = N$ the number of accessible states.

PROBLEM 4 — MIXED PARTIALS AND A FULL CLASSIFICATION

(a) Verify Clairaut's theorem explicitly for $f(x,y) = x^3y^2 + \mathrm{e}^{xy}$ by computing $\partial_y\partial_x f$ and $\partial_x\partial_y f$ separately. (b) Find all critical points of $h(x,y) = x^3 + y^3 - 3xy$ and classify each one using the eigenvalues of its Hessian, stating the principal curvature directions.

– Solution

(a) Treating y as constant, $\partial_x f = 3x^2y^2 + y\,\mathrm{e}^{xy}$. Now differentiate that in y, using the product rule on both terms:

$$\partial_y\partial_x f = 6x^2y + \mathrm{e}^{xy} + xy\,\mathrm{e}^{xy}.$$

The other order: $\partial_y f = 2x^3y + x\,\mathrm{e}^{xy}$, and differentiating in x,

$$\partial_x\partial_y f = 6x^2y + \mathrm{e}^{xy} + xy\,\mathrm{e}^{xy}.$$

Identical. ✓ Both second partials are continuous everywhere (they are polynomials and exponentials), so Clairaut's hypothesis holds and the agreement was guaranteed in advance.

(b) Critical points. $\partial_x h = 3x^2 - 3y = 0$ and $\partial_y h = 3y^2 - 3x = 0$ give $y = x^2$ and $x = y^2$. Substituting, $x = x^4$, so $x(x^3-1) = 0$ and $x \in \{0,1\}$ over the reals. The critical points are $(0,0)$ with $h = 0$, and $(1,1)$ with $h = 1+1-3 = -1$.

The Hessian. $H(x,y) = \begin{pmatrix} 6x & -3 \\ -3 & 6y \end{pmatrix}$.

At $(0,0)$**:** $H = \begin{pmatrix} 0 & -3 \\ -3 & 0 \end{pmatrix}$, characteristic equation $\mu^2 - 9 = 0$, so $\mu = \pm 3$. The eigenvector for $\mu = +3$ is $(1,-1)/\sqrt{2}$ and for $\mu = -3$ is $(1,1)/\sqrt{2}$ (check: $H(1,-1)^\mathsf{T} = (3,-3)^\mathsf{T}$ ✓). Mixed signs, so a **saddle**: h curves upward along $(1,-1)$ and downward along $(1,1)$, each with curvature 3.

At $(1,1)$**:** $H = \begin{pmatrix} 6 & -3 \\ -3 & 6 \end{pmatrix}$, eigenvalues 6 ∓ 3, i.e. $\mu = 3$ along $(1,1)/\sqrt{2}$ and $\mu = 9$ along $(1,-1)/\sqrt{2}$. Both positive, so a **local minimum**, three times stiffer across the diagonal than along it. The contours near it are ellipses elongated by $\sqrt{9/3} = \sqrt{3}$ in the $(1,1)$ direction.

Sanity checks worth doing. The determinant of H is $36xy - 9$, which is $-9 < 0$ at the origin and $+27 > 0$ at $(1,1)$ — and since $\det H = \mu_1\mu_2$ (Chapter 0.5), a negative determinant in two dimensions *immediately* signals opposite signs, hence a saddle, with no eigenvector computation at all. That is the standard "second derivative test" for two variables: $\det H < 0$ means saddle, $\det H > 0$ with $\partial_x^2 h > 0$ means minimum. It is not a separate rule — it is §6.3 plus the fact that the determinant is the product of the eigenvalues, and it stops being usable in three dimensions, where a positive determinant is compatible with two negative eigenvalues. The eigenvalue formulation is the one that generalises.

> Note also that h is unbounded below (take $y = 0, x \to -\infty$), so the local minimum at $(1, 1)$ is emphatically *not* global. §6 is a local theory: the Hessian is a statement about a neighbourhood and knows nothing about what happens far away.

THE BRICK YOU JUST LAID

You took one equation from Chapter 0.1 and let the displacement be a vector. Out of that came the **total derivative** $\mathrm{D}f(\mathbf{a})$ — a linear map, unique, whose matrix is the Jacobian — and the recognition that partial derivatives are its components and can exist without it. You derived the steepest-ascent property of the **gradient** from Cauchy–Schwarz and its perpendicularity to level sets from the chain rule; you saw that turning $\mathrm{d}f$ into ∇f requires a **metric**, and met your first covector doing it. The chain rule became a **product of Jacobians** and quietly produced the summation convention. The **Hessian**, symmetric by Clairaut and therefore diagonalisable by the spectral theorem, classified critical points by its eigenvalues and told you the principal curvatures. **Lagrange multipliers** came from a picture — level set tangent to constraint — and turned out to be sensitivities. And the **Jacobian determinant** converted local linearity plus the determinant-as-volume into every change-of-variables formula there is.

Where this gets spent.

- **The total derivative** → Chapter 1.2, where the same definition applied to a functional gives the Euler–Lagrange equations; Chapter 3.2, where $\mathrm{D}f$ becomes the pushforward between tangent spaces on a manifold.
- **The gradient as a one-form** → Chapter 2.4 (upper and lower indices, made a convention), Chapter 3.2 (the cotangent space), Chapter 3.3 (raising and lowering with $g^{\mu\nu}$), Chapter 3.5 (where $\mathrm{d}f$ becomes the exterior derivative and $\mathrm{d}^2 = 0$ starts doing work).
- **The chain rule as a matrix product** → Chapter 2.4, where (0.6.26), promoted from a computation to a *definition*, is what a tensor is; Chapter 3.2, where it is how you change charts on a manifold.
- **Hessian + spectral theorem** → Chapter 0.8 (normal modes: eigenvalues of the Hessian of the potential are the squared frequencies), Chapter 1.3 (stability of equilibria in phase space), Chapter 6.6 (the Higgs potential, where a zero eigenvalue is a massless particle).
- **Lagrange multipliers** → Chapters 1.2 and 1.3 (constraint forces are multipliers, and now you know why they have units of force: they are sensitivities of an energy to a displacement), and later the constrained-Hamiltonian treatment of gauge theories, where the multipliers enforce Gauss's law.
- **The Jacobian determinant** → Chapter 0.2's debt, now paid; Chapter 3.3's $\sqrt{-g}\,\mathrm{d}^4x$; Chapter 5.5, where changing variables inside a path integral produces a determinant that is not a nuisance but a new field.

One thing to carry above all: every result in this chapter came from replacing a number by a linear map. That is the only idea here. Chapter 0.7 replaces the linear map by something that varies from point to point and asks what happens when you integrate it over a surface — and gets Stokes' theorem, which is the fundamental theorem of calculus with the same substitution applied.